

Using USD in Pixar's Digital Backlot

Eliot Smyrl
Pixar Animation Studios
United States of America
eliot@pixar.com



Figure 1: A few hundred models out of more than 33,000 available (©Pixar)

ABSTRACT

At Pixar we have developed a set of tools to resurrect more than 33,000 previously-unusable set & prop models from our old films to be used as a studio-wide resource for previs, set dressing, cameos, automated testing, shader library development test subjects, short film and streaming projects, VR projects, and research. Based on the extensive use of USD [Disney/Pixar 2016], the Digital Backlot has now been used for four released feature films, three feature films still in production, and six completed short-form projects (with several more under development). Our pipeline also ensures that future asset development will continue to build up this library.

CCS CONCEPTS

- **Computing methodologies** → *Graphics systems and interfaces*;
- **Applied computing** → *Media arts*.

KEYWORDS

USD, scene description, feature film production, obsolescence, ptex, shader baking

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1 MOTIVATION

For many years, the rapid pace of pipeline development at Pixar made models from past shows technically obsolete very quickly, a problem faced by other studios as well. This often frustrated attempts to save production time by re-using past designs, particularly for sequels. Several previous attempts to collect a library of models became obsolete as quickly as the original assets.

However, several factors finally made a "backlot project" feasible and even necessary:

- The advent of USD [Disney/Pixar 2016] as a description of geometry and shader networks brought stability to our encoding scheme
- The production design of *Toy Story 4* presented a need for thousands of unique models in the background of many shots in the film in the "Antiques Mall" set, which drove us to look again at how we could use the many models we've built during almost 30 years of film production.
- The rapidly-increasing desire to produce short projects based on previously-created worlds demands cheap, simple ways to re-use assets.

2 APPROACH

The first set of decisions we made that enabled this project to succeed, when previous attempts had failed, was to reduce the scope of the problem in several ways.

First, we limited ourselves to including only set and prop models with no or very limited articulation. The majority of models created for a film fall into this category, and this eliminates a sizable set of issues related to maintaining complex character articulation developed through multiple complete rewrites of our articulation system.

Next, we limited the primary use case in the *destination* project to mid- and background models. This allows us to make simplifying decisions that don't maintain exact fidelity to the original look of the model, but are "close enough".

We also made some initial restrictions - namely a complete absence of rigging, and no procedural geometry - that we have since been able to relax to some extent. We now support limited affine-transformation rigging and UsdGeomPointInstancer primitives.

3 BENEFITS OF USD

We store the description of each model (geometry and shader network/binding encoding) in USD. Since USD is an open-source format, it's proven to be more stable than any of our past internal formats, for the same reason that the Maya file format has been stable: wide adoption is a damping function on change.

We have also relied on USD's layering/referencing architecture; this has allowed us flexibility in the file structure of the model (in particular to have different parts of the model installed in different filesystems) while externally presenting the same composed structure as one of our standard production models. For example, Backlot assets can be localized to particular productions' workspaces while retaining references to both the "canonical" assets and to global shading setups that allow broad adjustments as technology and needs evolve.

In addition, as industry adoption of USD spreads, it has led to novel use cases such as easy transport to game engines for prototypes of VR projects and for virtual production.

4 BAKED SHADING

The original shading information is "baked" out into textures. While this samples the original shading, thus limiting its use, this is acceptable given the use case (mid- to background). In practice, the models turned out to look better than we expected at a much larger range of view distances.

Our baking process creates textures for the inputs to the illumination model - diffuse and specular coefficients, for example. Given the variety of historical illumination models we were attempting to capture, and the inputs to the current illumination model to which we needed to map these old values, we have 43 channels of data (some of which are scalar values such as coefficients, some of which are vector values such as normal vectors, and some of which are colors such as reflectance tint).

We store this baked data as ptex [Burley and Lacewell 2008], and generate a simple standard USD shading network that uses this ptex data and applies it to illumination inputs. For different illumination models, we interpose a different adapter function between the baked

signals and the illumination inputs. Since we are not attempting an exact duplication of the original look, the adapter is empirical rather than rigorous.

5 USER INTERFACE

Since this library comprises a larger set of models than we've ever dealt with at one time before, we knew that organization would be very important. The models are divided into category groups organized the same way they were in their original show, which helps people who are looking for a specific model they remember from past work. In addition, the models have keyword tagging ("chair", "tool", "food", etc). A large original set of tags was originally generated and applied by users, who are in a position to best know what keyword sets are most useful to them, and we are now finalizing a machine-learning model to extend this tagging data throughout the library.

When users then examine a category, or a set of models with a keyword tag, they browse visual thumbnails using a scrolling grid of widgets. Each widget shows multiple thumbnail rendered views of each model, with mouse motion over the widget scrolling through the views. This enables very rapid decisions about which model the user would like, and how it will look from many angles.

Once the user settles on a model, a very simple single-click interface brings the model into their show and adds it directly to a set or a shot, including building all the behind-the-scenes infrastructure necessary to make that happen.

6 CONCLUSIONS

Our Digital Backlot has proved invaluable now for a variety of projects. It was used for over 40% of the set/prop models in *Cars 3* and reduced the startup time of the Layout department to days instead of months, since they could begin location scouting in Radiator Springs immediately without having to import or resurrect old models. *Toy Story 4* saw a similar scope of use. However, not all shows have been able to use it this successfully - *Coco* for example had a unique design sensibility that precluded the use of objects designed for other shows.

The Digital Backlot has been most successful for short projects, in which it has been used for up to 90% of set/prop models, saving scarce resources for use elsewhere and allowing artists to realize projects that would previously have been infeasible. It has also enabled exploratory pre-production creativity for upcoming features, since artists can grab "free" stand-in Backlot models and then later throw them away or update them as desired. There have also been a variety of new use cases, as mentioned earlier.

We have been expanding the Digital Backlot through a continuous process of adopting models from new productions (at a rate of approximately 3500 new models per year). Maintenance of existing models has been very inexpensive, validating the choice of USD as the "archive format".

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